

	$t - 20$	$t - 15$	$t - 10$	$t - 5$	t	causal	$\hat{\sigma}$
Senseiver-A						yes	current second only
Senseiver-G (ours)						yes	last 16 s
DINCAE						no	✓ one second either side
4DVarNet						no	fixed 200-s tile containing t
4DVarNet (aug. head)						no	✓ fixed 200-s tile containing t
EnKF						yes	✓ all past, carried by the ensemble

observations used reconstructed frame t